

Public Chat Portal for Alkhidmat

(AI-POWERED Q&A + HUMAN ROUTING)

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The Problem

Alkhidmat Foundation Pakistan serves millions through ~12 humanitarian domains – healthcare, disaster relief, education, orphan care, etc.

Current Challenges

1. Queries handled manually via phone and email.
2. Delays and repetitive responses overload staff.
3. Users with low digital literacy struggle to find information online.

Problem Statement:

“No centralized, intelligent chat solution exists to handle diverse user queries efficiently and inclusively.”

Our Solution: An AI-powered multilingual chat portal powered by a RAG

Automated FAQs, classifies queries by domain, and routes complex questions to staff.

Why this project matters:



**Enhances
accessibility for all,
especially
underserved
communities.**



**Provides
instant
responses to
queries, 24/7.**



**Reduces staff
workload,
improving
efficiency and
transparency.**



**Strengthens
public trust
through timely
communication.**

Existing solutions and Gaps

- Platforms like WHO HealthBot, NOOR AI, and FBR Chatbot provide public or social service support.
- Mostly rule-based or FAQ-driven, handling only one or two domains (e.g., health, legal aid, or registration).
- Often English-only or partially Urdu, excluding Roman Urdu and regional users.
- No AI-human escalation or context-aware conversation flow.
- No analytics or feedback insights to guide service improvement.

Novelty

- 12-Domain Integration: Covers all Alkhidmat services and sub-branches, handling complex, non-generic queries.
- Agentic AI Workflow: Multilingual (Urdu, Roman Urdu, English) agents with secure intent verification.
- Custom Knowledge Base: Context-aware RAG built from real Alkhidmat data—not generic sources.
- Human-AI Escalation: Smart routing to staff dashboard for empathy and efficiency.
- Advanced Analytics: Monitors trends and gaps for ongoing improvement.

Real-world Impact

Empowers citizens to easily connect with Alkhidmat's services (e.g., donations, medical aid, scholarships).

An application of **AI for Social Good** in Pakistan's humanitarian sector.

- Accessibility: 24/7 chat in Urdu + Roman Urdu breaks literacy barriers.
- Efficiency: Reduces staff load for repetitive questions.
- Transparency: Logs interactions; useful for reporting and audits.
- Scalability: Same model can be reused by other NGOs in Pakistan.
- Inclusivity: Designed for people with low digital literacy or rural access.

CS Rigor

Core Computer Science Areas Involved:

Natural Language Processing (NLP):

Intent detection & multilingual text understanding.

Information Retrieval:

Retrieval-Augmented Generation (RAG) for accurate answers.

Machine Learning & Deep Learning:

Lightweight classifiers & contextual response generation.

Human-Computer Interaction (HCI):

Accessible chat design for low-literacy users.

Database Systems:

Efficient indexing & data management using Elasticsearch.

Data Analytics:

Query trend analysis for service improvement.

Scope

Main Modules:

- Chatbot Interface (Frontend): Multilingual UI (English, Urdu, Roman Urdu)
- Backend API (FastAPI + LangChain): RAG pipeline for context-aware responses
- Intent Classifier (ML)
- Human Staff Dashboard (React)
- Analytics Module (Data Logs & Insights)

Semester 1 (Phase 1):

- Further Requirement gathering, domain mapping
- UX design, prototype chatbot, classifier setup

Semester 2 (Phase 2):

- Full integration & dashboard (RAG integration as well)
- Logging, analytics, testing, deployment and documentation

Feasibility

Data:

- Alkhidmat's website, reports, FAQs, and announcements (freely available).
- Supplementary open-source Q&A datasets for pretraining.

Resources:

- Compute: University machines + lightweight cloud hosting.
- Libraries: FastAPI, LangChain, PyTorch, scikit-learn, React, Elasticsearch.
- Hardware: Standard laptops (8–16GB RAM).

Cost & Scalability:

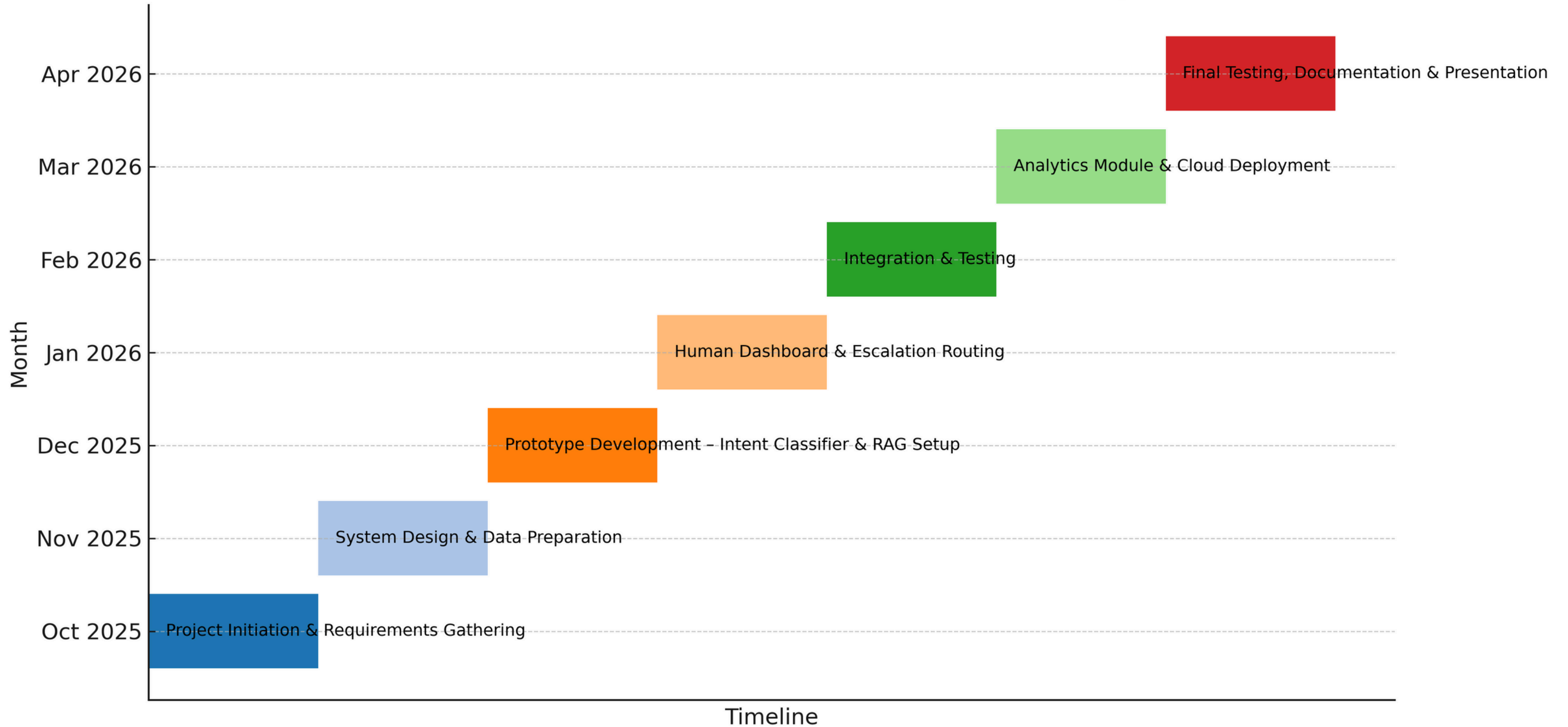
- Fully open-source stack, zero licensing cost.
- Cloud deployment via Docker → easily scalable.

Team Dynamics & Responsibilities Overview

Member	Relevant Courses	Primary Role	Key Responsibilities
Fatima	- Modelling Social Complexity - Software Engineering - Reinforcement Learning	AI & Frontend Integration Lead	- Chatbot UX & multilingual flow design - Agent orchestration - AI-UI integration
Yashfeen	- Deep Learning - Data Science	Machine Learning Engineer	- Intent classifier training - Confidence estimation for routing - Evaluation metrics & ML tuning
Rubab	- Web & App Development - Data Science	Analytics & Dashboard Lead	- Data logging pipeline for user queries - Staff dashboard visualizations & reports - Insights on domain-wise trends & unmet needs
Samah	- Enterprise Software Development - Software Engineering - Data Science/ Deep learning	Backend & Deployment Lead	- API & database design - RAG & intent routing implementation - Cloud deployment & testing

End of project							
	YEAR 1			YEAR 2			
	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	March 2026	April 2026
	Requirements approval	Design completion	Prototype ready	Human routing operational	Full integration	Deployed beta	Final defence
Phase / Task	Phase 1 – Project Initiation & Requirements Gathering	System Design & Data Preparation	Prototype Development – Intent Classifier & RAG Setup	Phase 2 – Human Dashboard & Escalation Routing	Integration & Testing	Analytics Module & Cloud Deployment	Final Testing, Documentation & Presentation
Deliverables & Milestones	- Requirements document - Domain map (12 Alkhidmat domains) - User flow & chatbot conversation map	- UX wireframes for chat interface - Dataset collection & cleaning from Alkhidmat resources - Initial ER diagram and API routes	- Trained ML classifier (intent detection) - Integrated RAG pipeline using LangChain + Llama 2 - Basic chatbot interface connected to backend	- React dashboard for staff queries - Escalation logic & workflow integration - Internal test report (AI ↔ Human routing)	- Full system integration (chat portal + dashboard + RAG) - Unit & integration tests - Initial usability feedback	- Analytics dashboard (query trends, escalation rates) - Dockerized deployment (Heroku) - Beta version live for demo	- Final project report & documentation - User guide & demo video - Final FYP presentation and viva
Responsible Members	All members (collaboration) Lead: Samah (documentation structure), Fatima (domain mapping)	1.Fatima: UX wireframes 2.Samah: Database schema & API design 3.Yashfeen: Data preprocessing 4.Rubab: Data logging plan	1.Yashfeen: Classifier model 2.Fatima: Frontend integration 3.Samah: Backend FastAPI endpoints 4.Rubab: Data logs for test runs	1.Samah: Backend logic & dashboard APIs 2.Rubab: Dashboard UI/UX 3.Fatima: Chatbot ↔ dashboard connection 4.Yashfeen: Confidence-based routing model	1.Samah: Backend & deployment tests 2.Yashfeen: Model evaluation 3.Rubab: Usability test reports 4.Fatima: Frontend bug fixes	1.Rubab: Analytics module 2.Samah: Docker + Heroku deployment 3. Fatima: UI enhancements 4.Yashfeen: Data validation metrics	All members (joint effort) 1.Samah: Technical report 2.Rubab: Visuals & analytics 3.Fatima: Presentation slides 4.Yashfeen: Model performance summary
Deliverable Output	Project charter, domain map, workflow	UX wireframes, DB schema	Basic AI chatbot working	Dashboard connected	Working AI + Human chat flow	Live Dockerized version	Documentation & presentation

Gantt Chart - Public Chat Portal for Alkhidmat (Oct 2025 - Apr 2026)



Thank You

FOR YOUR TIME